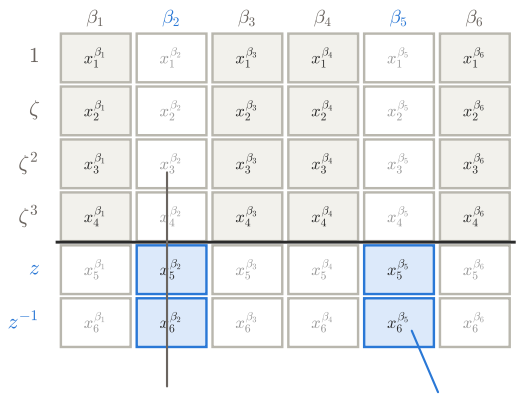


one term per choice of the two free columns $U = \{j_1 < j_2\}$ and every choice falls into one of three types



$$M_S = (-1)^{\text{inv}(b_S)} V$$

Lemma 4.1: zero unless the t residues on S are distinct

$f(\beta_{j_1} - \beta_{j_2})$
the free 2×2 block



a residue class is empty
profile (3, 2, 1, 0)
0 terms



two classes of size two
profile (2, 2, 1, 1)
4 terms



one class of size three
profile (3, 1, 1, 1)
3 terms

● residue 0 ● residue 1 ● residue 2 ● residue 3